

EDUCATION

•National Institute of Technology, Kurukshetra

BTech: Information Technology

2022-2026

CGPA: 8.25/10

EXPERIENCE

•Software Developer (Freelance)

Rajasthan Technical University, Kota | [View Project](#) | [View Github](#)

Dec 2024 – Jun 2025

- Built a scalable MERN platform serving 100+ active users, implementing controller-service architecture for modular backend design
- Designed backend using DTO pattern to ensure structured and efficient data transfer between layers
- Designed and deployed role-based task workflow system, handling assignments, and evaluation across 3+ user roles
- Implemented real-time notifications and leaderboard using WebSockets, improving user engagement by 35%
- Integrated face recognition-based attendance system, reducing manual verification effort by 80%
- Developed admin dashboard for analytics, tracking user activity, task completion rates, and engagement metrics
- Automated background jobs (email notifications, scheduling), reducing manual intervention by 60%
- Optimized backend APIs and database queries, improving response time by 30–40%

PROJECTS

•Code Collaborator | [View Project](#) | [View Github](#)

MERN, WebSockets, Monaco Editor

- Built a **real-time collaborative coding platform** using **WebSockets**, supporting **5–10 users** per room with instant join/leave notifications and live code sync
- Engineered **room-based architecture** with shareable links, **4-language** support, and zero-conflict state management across all connected clients
- Integrated a **shared whiteboard** with the code editor enabling real-time diagramming, ideal for **technical interviews** and pair programming
- Secured sessions via **JWT authentication** with authorized room access and **graceful disconnection handling** to auto-cleanup rooms on user exit

•Real-Time Peer-to-Peer Video Calling Platform

WebRTC, WebSockets, Node.js, React

[View Github](#)

- Built a peer-to-peer video calling application using WebRTC enabling direct browser-to-browser communication
- Designed custom signaling server using Socket.io for offer/answer exchange and connection setup
- Implemented real-time audio/video streaming with low latency (sub-second delay)
- Developed room-based system for 1-to-1 secure video sessions
- Integrated ICE candidate exchange for stable peer connections across different networks
- Optimized connection setup time and improved call stability through efficient signaling handling

TECHNICAL SKILLS

Languages: C/C++, JavaScript, SQL

Backend: Node.js, Express.js, WebSockets

Frontend: React.js, CSS, Ant Design, HTML

Databases: MongoDB

Cloud: MongoDB Atlas

Real-time: WebRTC, Socket.io

Backend Architecture: Controller-Service Pattern, REST APIs, DTO Design, Scalable Systems

Core: DSA , OS, OOPS, DBMS

ACHIEVEMENTS

- **Coding Profile** : Built and deployed multiple production-ready full-stack applications used by real users
- **Smart India Hackathon** : College Winner, National Nominee 2024
- **Excalibur** : Top 7 / 60 teams 2023
- **Flipkart Grid 7.0** : Semi-finalist 2025
- **Thought Lab Club** : Student Secretary 2025